



Graphene can be 3D printed

Researchers at Virginia have found a novel technique to 3D print graphene, the hardest material on earth. <https://digit.in/oct18-77-1>



Sound to print droplets

Researchers have developed printing tech that uses sound waves to print droplets independent of fluid viscosity. <https://digit.in/oct18-77-2>

by difficulty in social skills, repetitive behaviour, speech and nonverbal communication. There's not just one type of autism, and the way it manifests in someone depends on the sub-type that affects them. They include autistic disorder, childhood disintegrative disorder, pervasive developmental disorder-not otherwise specified (PDD-NOS) and Asperger syndrome. As a result, treatments for autism often target one particular diagnoses instead of the entire spectrum.

THE SCIENCE

Fragile X syndrome is one of the most common causes of autism spectrum disorder. Researchers have been able to use CRISPR-GOLD on mice to successfully edit the gene for a neurotransmitter. The method, which employs gold particles to deliver Cas9 to the brain, was developed at the University of California, Berkeley. The result? Reduction in the repetitive behaviour caused by the fragile X syndrome (FXS).

Why target the fragile X syndrome? Broadly, autism spectrum disorders (ASDs) are complex from a genetic perspective, whereas single gene disorders like the fragile X syndrome are easier to target. However, that also makes it rare, occurring in only one in every 4,000 boys and 6,000 girls.

The GOLD method has its own advantages. Usually, scientists are able to deliver the Cas9 protein via a virus into neurons. The problem with this method is the continuous expression of the Cas9 enzyme, which might lead to unintentional editing of other genes. The CRISPR-Gold method avoids this by controlling the amount that needs to be inserted, and carries it safely to the target site without contamination or unintended expression.

During this experiment, researchers injected CRISPR-GOLD with the Cas9 complex into the striatum of the brain – the part of the brain that handles habitual behaviour – and targeted metabotropic glutamate receptor 5 (mGluR5), an excitatory receptor that plays an important role

in inter-neuron communication and is affected adversely by FXS. After disabling the gene responsible for mGluR5, the repetitive behaviour in mice was noticeably reduced. In mice suffering from the syndrome, the behaviour in question was persistent digging and erratic leaps into the air. As a result of the treatment, the former reduced by 30 percent while the latter was brought down by 70 percent.

THE ETHICS

Once the science of this research is understood, it brings us to the other side of this debate - the ethics. Autism activists and organisations across the world have been trying to make people perceive autism not as a disease or a

tistics, such as those who have Asperger's syndrome. There are other forms of autism which result in direct physical consequences on one's health. Many individuals with more severe forms of autism are prone to chronic digestive problems, intellectual disabilities, sensory sensitivities, severe anxiety, sleep disorders, self-injurious behaviour, and more. Quite obviously, this does not allow them to lead a regular life.

There are studies that have revealed that people with autism have a higher tendency to suffer from obesity, depression and even die 16 years earlier than a non-autistic person on an average. While some of these reasons are directly affected by the social stigma associated with autism, others are

not. Not to mention the costs that families have to bear to afford suitable care and caretakers, whether or not they are financially capable of doing so.

There is no denying that a lot more needs to be done to free the society from the myths and stigmas associated to autism. An openness to neurodiversity, appropriate counselling along with affordable, available caretaking facilities could go a long way. However, this also makes it necessary that work towards

accessible cures, if and when possible, is equally necessary for individuals affected, directly or indirectly. From where we stand today, CRISPR-Cas9 might open the doors to a day where autistic individuals can make the choice of living a full life with their condition, curing aspects of it or getting rid of it altogether. Opposing the mere possibility of that day is not just foolhardy, it is downright detrimental to the evolution of choice available to autistic individuals today. 



Students and family members from Johnson Primary School march, holding signs and banners in support of autism awareness aboard Marine Corps Base Camp Lejeune, Friday. According to the center for disease control and prevention, one in every 68 children in the U.S. is diagnosed with the disorder.

disorder but as a different way of living. It is not just an opinion prevalent among a group of people, however. There's been research to prove that autistic people are sometimes better problem solvers (digit.in/Aut1) and more creative (digit.in/Aut2). According to this side of the argument, editing out autism from the human race would count as a genetic cleansing of sorts and span into eugenics territory.

However, much of the debate on this side is focussed on high-functioning au-